The bounded cylindric polynomials $Q_{n,c}(q)$ for $d = 2$: a Rogers–Ramanujan mechanism

Seed 1, Layer 2, Round 2

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Abstract

For $r = 3$ and $d = 2$ we prove $Q_{n,c}(q) = q^{n^2}$ for all profiles c in the cyclic orbit of $(1, 1, 0)$, and $Q_{n,c}(q) = q^{n(n+1)}$ for all c in the orbit of $(2, 0, 0)$. In particular Warnaar's positivity conjecture holds for $d = 2$. The proof runs through a general lemma, valid for every profile and rank: multiplying the Corteel–Welsh functional equation by $(yq; q)_\infty$ converts it into a q-difference system with *polynomial* coefficients $(-1)^{|J|-1}(yq; q)_{|J|-1}$. For $d = 2$ this system collapses to the classical Rogers–Ramanujan q-difference equation $G(y) = G(yq) + yq\,G(yq^2)$, whose coefficients are manifestly nonnegative; this is precisely why the alternating signs of $(yq; q)_\infty$ disappear.

1 Setting

Fix $r = 3$. For a profile $c = (c_1, c_2, c_3)$ with $d = c_1 + c_2 + c_3$, let $F_c(y, q) = \sum_{\Lambda\in\mathcal{C}_c} q^{|\Lambda|}y^{\max(\Lambda)}$ be the bivariate cylindric partition generating function, and write $F_c(y, q) = \sum_{n\geq 0} f_{c,n}(q)\, y^n$. For fixed n and w the number of cylindric partitions with $\max(\Lambda) = n$ and $|\Lambda| = w$ is finite (all parts are $\leq n$, so the total size w bounds the number of parts); hence each $f_{c,n} \in \mathbb{Z}[[q]]$ and $F_c \in \mathbb{Z}[[q]][[y]]$. All identities below are identities in the ring $\mathbb{Z}[[q]][[y]]$, in which any element with constant term 1 (e.g. $1 - yq^j$, $1 - q^n$, $(yq; q)_\infty$) is a unit.

For $d = 2$ we have $\ell = \gcd(d, 3) = 1$, so

$$Q_{n,c}(q) = (q; q)_n\, [y^n]\big((yq; q)_\infty\, F_c(y, q)\big).$$

Since rotating the profile rotates the cylinder — a bijection preserving $|\Lambda|$ and $\max(\Lambda)$ — F_c, and hence $Q_{n,c}$, depends only on the cyclic orbit of c.

For $d = 2$ there are exactly two orbits, represented by $a = (1,1,0)$ and $b = (2,0,0)$. Write $A(y) = F_{(1,1,0)}(y,q)$ and $B(y) = F_{(2,0,0)}(y,q)$.

Our tool is the Corteel–Welsh functional equation: for any profile c, with $I_c = \{i : c_i > 0\}$ and $c(J)$ the shifted profile,

$$F_c(y,q) = \sum_{\emptyset \subsetneq J \subseteq I_c} (-1)^{|J|-1} \frac{F_{c(J)}(yq^{|J|}, q)}{1 - yq^{|J|}}. \tag{1}$$

2 The key lemma: where the alternating signs go

Lemma 1 (G-transform of the Corteel–Welsh system). *For any rank r and any profile c, set $G_c(y) := (yq;q)_\infty \, F_c(y,q)$. Then*

$$G_c(y) = \sum_{\emptyset \subsetneq J \subseteq I_c} (-1)^{|J|-1} \, (yq;q)_{|J|-1} \, G_{c(J)}(yq^{|J|}). \tag{2}$$

Proof. Substitute $F_{c(J)}(yq^{|J|}, q) = G_{c(J)}(yq^{|J|})/(yq^{|J|+1};q)_\infty$ into (1) and multiply both sides by $(yq;q)_\infty$. The coefficient of $G_{c(J)}(yq^{|J|})$ becomes, with $s = |J|$,

$$\frac{(yq;q)_\infty}{(1 - yq^s)\,(yq^{s+1};q)_\infty} = \frac{(yq;q)_\infty}{(yq^s;q)_\infty} = (yq;q)_{s-1},$$

using $(yq^s;q)_\infty = (1 - yq^s)(yq^{s+1};q)_\infty$ and $(yq;q)_\infty = (yq;q)_{s-1}(yq^s;q)_\infty$. □

Remark 1. *Lemma 1 explains the role of the sign-alternating factor $(yq;q)_\infty$ in the definition of $Q_{n,c}$: it is exactly the cofactor that clears every denominator $1 - yq^{|J|}$ from the Corteel–Welsh system, leaving a q-difference system with* finite polynomial *coefficients $(yq;q)_{|J|-1}$. Since $Q_{n,c} = (q^\ell;q^\ell)_n \, [y^n] G_c(y)$, the positivity conjecture is a statement about the solution of the polynomial-coefficient system* (2)*.*

3 The case $d = 2$

Theorem 1. *For all $n \geq 0$,*

$$Q_{n,c}(q) = q^{n^2} \quad \text{for } c \in \{(1,1,0),(0,1,1),(1,0,1)\}, \qquad Q_{n,c}(q) = q^{n(n+1)} \quad \text{for } c \in \{(2,0,0),(0,2,0)$$

In particular the Corteel–Dousse–Uncu/Warnaar positivity conjecture holds for $d = 2$.

Proof. Step 1: the system. For $c = (1,1,0)$ we have $I_c = \{1,2\}$ and (indices cyclic, $c_0 = c_3$): $c(\{1\}) = (0,2,0)$, $c(\{2\}) = (1,0,1)$, $c(\{1,2\}) = (0,1,1)$. Using cyclic invariance to replace $F_{(0,2,0)}$ by B and $F_{(1,0,1)}, F_{(0,1,1)}$ by A, (1) reads

$$A(y) = \frac{B(yq)}{1-yq} + \frac{A(yq)}{1-yq} - \frac{A(yq^2)}{1-yq^2}. \tag{3}$$

For $c = (2,0,0)$ we have $I_c = \{1\}$ and $c(\{1\}) = (1,1,0)$, so

$$B(y) = \frac{A(yq)}{1-yq}. \tag{4}$$

Step 2: eliminate B. Substituting (4) (at $y \mapsto yq$) into (3),

$$A(y) = \frac{A(yq)}{1-yq} + A(yq^2)\left[\frac{1}{(1-yq)(1-yq^2)} - \frac{1}{1-yq^2}\right] = \frac{A(yq)}{1-yq} + \frac{yq\,A(yq^2)}{(1-yq)(1-yq^2)},$$

since $1-(1-yq) = yq$. Multiplying by $1-yq$:

$$(1-yq)\,A(y) = A(yq) + \frac{yq}{1-yq^2}\,A(yq^2). \tag{5}$$

Step 3: the G-transform. Let $G(y) = (yq;q)_\infty A(y)$. Substituting $A(y) = G(y)/(yq;q)_\infty$, $A(yq) = G(yq)/(yq^2;q)_\infty$, $A(yq^2) = G(yq^2)/(yq^3;q)_\infty$ into (5) and multiplying by $(yq;q)_\infty = (1-yq)(yq^2;q)_\infty = (1-yq)(1-yq^2)(yq^3;q)_\infty$ gives

$$(1-yq)\,G(y) = (1-yq)\,G(yq) + yq\,(1-yq)\,G(yq^2).$$

Dividing by the unit $1-yq$:

$$G(y) = G(yq) + yq\,G(yq^2). \tag{6}$$

This is the classical Rogers–Ramanujan q-difference equation. (Equivalently: apply Lemma 1 to both profiles — the $|J| = 1$ terms carry coefficient $+1$, the single $|J| = 2$ term carries $-(1-yq)$, and elimination of G_B recombines the signs into the nonnegative coefficients 1 and yq of (6).)

Step 4: uniqueness and solution. Write $G(y) = \sum_{n\ge 0} g_n(q) y^n$. Comparing coefficients of y^n in (6):

$$g_n = q^n g_n + q^{2(n-1)} \cdot q\,g_{n-1} \qquad \Longleftrightarrow \qquad (1-q^n)\,g_n = q^{2n-1} g_{n-1} \quad (n \ge 1).$$

Since $1-q^n$ is a unit of $\mathbb{Z}[[q]]$, the g_n are uniquely determined by g_0. Now $g_0 = G(0) = F_c(0,q) = 1$ (only the empty cylindric partition has maximum 0), and by induction, using $\sum_{j=1}^n (2j-1) = n^2$,

$$g_n = \frac{q^{n^2}}{(q;q)_n}.$$

Therefore

$$Q_{n,(1,1,0)}(q) = (q;q)_n\, g_n = q^{n^2}.$$

Step 5: the orbit of $(2,0,0)$. Let $G_B(y) = (yq;q)_\infty B(y)$. By (4),

$$G_B(y) = \frac{(yq;q)_\infty}{1-yq}\, A(yq) = (yq^2;q)_\infty A(yq) = G(yq),$$

so $[y^n]G_B = q^n g_n = q^{n(n+1)}/(q;q)_n$ and $Q_{n,(2,0,0)}(q) = q^{n(n+1)}$.

By cyclic invariance the same formulas hold across each orbit. Both are monomials, hence have nonnegative coefficients; and $Q_{n,c}(1) = 1 = \big(\frac{1}{6}(d+1)(d+2)-1\big)^n$ for $d=2$, consistent with Welsh's evaluation. □

Remark 2 (Example)**.** *$n=2$, $c=(1,1,0)$: the recursion gives $g_1 = q/(1-q)$, $g_2 = q^3 g_1/(1-q^2) = q^4/\big((1-q)(1-q^2)\big)$, so $Q_2 = (q;q)_2\, g_2 = q^4$. Direct expansion of $(q;q)_2\,[y^2]\big((yq;q)_\infty F_{(1,1,0)}(y,q)\big)$, with $f_{c,0}=1$, $f_{c,1} = 2q/(1-q)$, $f_{c,2} = q^2(2+q+q^2)/\big((1-q)^2(1+q)\big) = 2q^2+3q^3+6q^4+7q^5+10q^6+\cdots$ (machine-verified against the Corteel–Welsh system), reproduces q^4 after the massive cancellation — verified to precision q^{900}.*

Remark 3 (Why the signs collapse, and what is special about $d=2$)**.** *Three separate phenomena combine: (i)* globally*, Lemma 1 absorbs the infinite alternating series $(yq;q)_\infty$ into finite coefficients $(yq;q)_{|J|-1}$; (ii)* for $d=2$, *elimination of the second orbit recombines the residual signs $\{+1,+1,-(1-yq)\}$ into the manifestly nonnegative coefficients $\{1,yq\}$ of* (6)*; (iii) the resulting recursion $(1-q^n)g_n = q^{2n-1}g_{n-1}$ is first order with monomial coefficient, which is why Q_n degenerates to a single monomial rather than a positive polynomial. For $d>2$, step (iii) genuinely fails (the eliminated system has higher order), but steps (i)–(ii) formulate the conjecture as a positivity property of the polynomial-coefficient system* (2)*; finding the general-d analogue of the sign recombination (ii) is the open problem.*

Remark 4 (Relation to known results)**.** *$t=k+d=5$ is the Rogers–Ramanujan modulus; equation* (6) *and its solution are classical (Rogers 1894), and Corteel–Welsh (2019) used* (1) *at $d=2$ to reprove the Rogers–Ramanujan identities. Warnaar (2023) proved positivity for $d=2$ via explicit multisums. The content here is the clean statement and proof in the $Q_{n,c}$ normalization, together with Lemma 1, which is profile- and rank-independent and isolates exactly where the alternating signs go.*

Verification

Script `scratch/scripts/seed1_R2L2_d2_verify.sage`: (1) brute-force enumeration of all cylindric partitions of size ≤ 25 for all six profiles matches the Corteel–Welsh solution (validating (3), (4) and the shifted-profile computations); (2) $Q_n = q^{n^2}$ and $Q_n = q^{n(n+1)}$ verified for $n \leq 10$ at precision q^{900} (well above the truncation-safety bound $6n^2 + 200 = 800$); (3) equations (5), (6), (2), $G_B(y) = G(yq)$ and $g_n = q^{n^2}/(q;q)_n$ each verified coefficientwise. All checks passed.